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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - April 2021

EMT618S – OPERATIONS RESEARCH

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. What is Operations Research?
2. Define degenerate basic feasible solution for a transportation problem.
3. Describe mathematical formulation of an assignment problem.
4. Define saddle point of a payoff matrix.
5. What are the main assumptions under in PERT computations?

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Solve graphically the following linear programming problem.
   Max Z = $2X_1 + 3X_2$
   Subject to : $X_1 - X_2 \leq 2$
   $X_1 + X_2 \geq 4$ and $X_1, X_2 \geq 0$

7. Determine an initial basic feasible solution to the following transportation problem using N.W.C.R.

	D_1	D_2	D_3	D_4	Supply
O_1	8	4	1	5	14
O_2	8	9	2	7	16
O_3	4	3	6	2	5
Required	6	10	15	4	35

8. Assign three jobs on three machines for following cost matrix.

Jobs	Machines		
	M1	M2	M3
J1	Rs.14	Rs.12	Rs.16
J2	Rs.11	Rs.17	Rs.21
J3	Rs.20	Rs.8	Rs.7

9. For what value of λ, the game with the following matrix is strictly determinable.

$$\begin{array}{c} \\ A_1 \\ A_2 \\ A_3 \end{array} \begin{array}{c} \begin{array}{ccc} player\ B & & \end{array} \\ \begin{pmatrix} \lambda & 6 & 2 \\ 1 & \lambda & 7 \\ -2 & 4 & \lambda \end{pmatrix} \end{array} \quad player\ A$$

10. Differentiate between PERT and CPM.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Use simplex method to

Maximize $z = 10x_1 + x_2 + 2x_3$

Subject to the constrains

$x_1 + x_2 - 2x_3 \leq 10$

$4x_1 + x_2 + x_3 \leq 20$

$x_1, x_2, x_3 \geq 0$

(or)

b) Use Big – M method to solve the following LPP.

Maximize $z = 6x_1 + 4x_2$

Subject to the constraints:

$2x_1 + 3x_2 \leq 30$

$3x_1 + 2x_2 \leq 24$

$x_1 + x_2 \geq 3$ and

$x_1 \geq 0, x_2 \geq 0$

12. a) Obtain an optimal solution for the following transportation problem.

7	3	2	2
2	1	3	3
3	4	6	5
4	1	5	

(or)

b) Use Vogel's Approximation method to obtain an initial basic feasible solution of the transportation Problem.

	D	E	F	G	Available
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
Demand	200	225	275	250	

13. a) Four different jobs can be done on four different machines and take down tie costs are prohibitively high for change overs. The matrix below gives the cost in rupees of producing job i on machine j:

Jobs	**Machine**			
	M_1	M_2	M_3	M_4
I_1	5	7	11	6
I_2	8	5	9	6
I_3	4	7	10	7
I_4	10	4	8	3

How should the jobs be assigned to the various machine so that the total cost is minimized.

(or)

b) Six jobs A,B,C,D,E and F have arrived at one time to be processed on a single machine. Assuming that no new jobs arrive thereafter, determine.

Job	A	B	C	D	E	F
Processing time (minutes)	7	6	8	4	3	5

i) Optimal sequence as per SPT rule ii) Completion times of the jobs
iii) Mean flow time iv) average in-process inventory

14. a) Solve the game theory whose pay-off matrix given by

Player B

$$\text{Player A} \quad \begin{matrix} \\ A_1 \\ A_2 \\ A_3 \end{matrix} \begin{matrix} B_1 & B_2 & B_3 \\ \end{matrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & -4 & -3 \\ 1 & 5 & -1 \end{pmatrix}$$

b) Solve the following game by using the principle of dominance.

Player B

Player A	I	II	III	IV	V	VI
1	4	2	0	2	1	1
2	4	3	1	3	2	2
3	4	3	7	-5	1	2
4	4	3	4	-1	2	2
5	4	3	3	-2	2	2

15. a) The time estimates (in weeks) for the activities of a PERT network are given below.

Activity	t_0	t_m	t_p
1 – 2	1	1	7
1 - 3	1	4	7
1 - 4	2	2	8
2 - 5	1	1	1
3 - 5	2	5	14
4 - 6	2	5	8
5 - 6	3	6	15

a) Draw the project network and identify all the paths through it.
b) Determine the expected project length
c) Calculate the standard deviation and variance of the project length
d) What is the probability that the project will be completed.
i) At least 4 weeks later than excepted time?
ii) No more than 4 weeks later than expected time?
e) If the project due date is 19 weeks, what is the probability of not meeting the due date?
f) The probability that the project will be completed on schedule if the scheduled completion time is 20 weeks.
g) What should be the scheduled completion time for the probability of completion to be 90%?

(or)

b) Tasks A,B,C,… H, I constitute a project, The precedence relationship are
$A < D; A < E; B < F; D < F; C < G; C < H; F < I; G < I.$
Draw a network to represent the project and find the minimum time of completion of the project when time, in days of each task is as follows;

Task:	A	B	C	D	E	F	G	H	I
Time:	8	10	8	10	16	17	18	14	9

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